

Understanding Border Control Technologies -Towards Decolonial and Responsible Constructs For Immigrant Populations

Arshia Dutta

Center for Doctoral Training, Department of Electronic Physical and Mathematical Sciences
Royal Holloway University of London
London, England, United Kingdom
arshia.dutta.2023@live.rhul.ac.uk
Arshia.Dutta.2023@live.rhul.ac.uk

Abstract

This study will investigate ethical and privacy concerns surrounding the increasing use of border screening technologies, such as biometric identification and emotion recognition, in the UK. While these technologies aim to enhance security, they often perpetuate systemic biases, disproportionately affecting immigrant communities from non-Western and marginalized regions. By conducting semi-structured interviews with immigrants and volunteers from London-based civil society organizations, the research will capture personal experiences and highlight the ethical challenges of these surveillance practices. The study will also include interviews with border personnel, developers, and security practitioners responsible for designing and implementing these technologies to shed light on the complex intersection of policy, technology design, and execution at the border, revealing how exclusionary frameworks are embedded in these systems. Finally, the study aims to develop a decolonial, responsible technology framework, rethinking the design and use of border control mechanisms for their just and inclusive organization.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Immigration, Border Control Technologies, Biometric Identification, Privacy, Security, Decolonial, Well-Being, Transparency, Accountability

ACM Reference Format:

Arshia Dutta. 2025. Understanding Border Control Technologies -Towards Decolonial and Responsible Constructs For Immigrant Populations. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3706599.3707601>

1 INTRODUCTION

The UK has increasingly turned to advanced AI-driven border technologies such as biometric identification and emotion recognition

systems to manage immigration and strengthen national security. These technologies, while designed to enhance the efficiency and reliability of border control, raise significant ethical and privacy concerns, particularly for immigrants from resource-constrained and non-Western communities [19]. The use of such systems is often seen as a way to bolster security; however, they also perpetuate systemic biases and existing inequalities in immigration procedures [2]. Technologies like iBorderCtrl, which assess the credibility of immigrants using biometric and emotion recognition tools, have been shown to misclassify individuals, leading to unfair outcomes, particularly for vulnerable groups [13]. These misclassifications result from the limitations of AI systems, which are often trained on datasets that do not reflect the diversity of the populations they are meant to serve, leading to discriminatory decisions against those from marginalized communities. Predictive analytics and risk evaluation tools, widely used to assess migrants' likelihood of compliance or risk, further exacerbate these issues by labeling certain individuals as "high risk" based solely on data interpretations [6]. This process lacks transparency and accountability, raising serious concerns about the misuse and potential abuse of sensitive personal data [10].

For immigrants, particularly asylum seekers fleeing persecution or conflict, the extensive surveillance and biometric data collection create a sense of constant monitoring and loss of personal autonomy, contributing to psychological stress and fear [21]. This stress is often compounded by the dehumanizing nature of the technologies, which reduce individuals to biometric data points rather than acknowledging their personal narratives and unique experiences [17]. The deployment of such systems reflects broader historical and systemic inequalities. Many of these technologies are developed and implemented within frameworks that reinforce colonial legacies, where the experiences and realities of immigrants from the Global South are marginalized [9]. The design and application of these technologies are often embedded with Western-centric views of security and risk, which fail to account for the complex social, political, and cultural contexts of immigrants arriving from non-Western countries [16]. This perpetuates a cycle of exclusion where immigrants, particularly those from marginalized regions, are disproportionately targeted and subjected to invasive forms of surveillance and control. This study seeks to address these ethical and privacy concerns by exploring the perspectives of those most affected by these technologies—immigrants themselves—as well as the border personnel and developers responsible for their implementation. Through semi-structured interviews with immigrants and caseworkers at a London-based charity, the research captures

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, April 26–May 01, 2025, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3707601>

the lived experiences of those subjected to AI-driven border screening technologies. Additionally, by engaging with the developers and security practitioners who design and implement these systems, the study aims to uncover the exclusionary design choices that reinforce biases and inequalities. Ultimately, this research advocates for a decolonial approach to border technology design, one that places human dignity, transparency, and accountability at the forefront of technological development. By acknowledging the lived realities of immigrants and recognizing the roles and responsibilities of those creating these systems, the study proposes a model of responsibility-based technology that is more inclusive and just. This approach seeks to move away from technologies that dehumanize and exclude, toward systems that respect and prioritize the well-being of immigrants, challenging the status quo of border surveillance and control.

- **RQ1.** How do immigrant populations in western contexts perceive border control and protection technologies?
- **RQ2.** How do security practitioners and border protection personnel design, develop and implement technologies for immigration purposes?
- **RQ3.** How can a decolonial and responsibility oriented approach towards border control technologies be developed?

2 RELATED WORK

In the section below, I highlight the related research in human computer interaction, responsible technology development and people centred privacy and security.

2.1 Historical evolution of border technologies

The evolution of border screening technologies marks a shift from manual processes to sophisticated biometric systems, driven by global security and migration trends. Initially, border controls relied on basic identity checks and manual inspections, which were inconsistent and prone to human error [1]. These early methods, focused on national security, lacked standardization, resulting in varied practices among border officials, often leading to arbitrary outcomes (Saunders, 2023). After World War II, increasing global mobility necessitated more reliable methods, leading to the widespread use of passports and visas as standardized identity tools [11]. However, the reliance on physical documents remained problematic due to forgery risks and time-consuming manual inspections [9]. The late 20th and early 21st centuries saw a pivotal shift with the integration of biometric technologies, especially following the 9/11 attacks [6]. Biometrics, including fingerprints, facial recognition, and iris scans, offered a more secure way to verify identities compared to traditional documents [23]. The UK adopted biometric passports and automated e-gates, streamlining border processes and reducing manual errors [3]. These technologies improved security and efficiency, addressing the challenges posed by rising international travel and the complexity of migration [17]. Despite setbacks in projects like the UK's e-borders, these systems continue to form the backbone of modern border management, reflecting the need for advanced solutions to address security and migration concerns [4, 24]. As technologies evolve, they mirror the ongoing need to balance security with efficient processing, responding to

both global mobility challenges and heightened security threats [16].

2.2 Role of technology in shaping social and political dynamics at the borders

The rise of advanced border screening technologies has significantly altered social dynamics at international borders, impacting how migrants are perceived and treated globally [18]. These technologies, such as biometric systems and AI-powered analytics, have intensified surveillance, often framing migrants as security threats rather than individuals seeking safety or opportunity [22]. This shift is deeply tied to the militarization of border management, where state security is prioritized over individual rights [11, 14]. Increased scrutiny reinforces stereotypes, particularly against racialized migrants who are disproportionately targeted by surveillance [22]. Tools like drones are used to monitor and manage migrants, exacerbating their dehumanization and complicating ethical border practices [22]. These technologies influence societal views on immigration, contributing to a hostile environment and marginalizing vulnerable groups [11]. The framing of migrants as threats shifts public attitudes, undermining efforts for inclusive immigration policies and raising concerns about justice and human rights [14, 20].

2.3 Technologies, policies, and immigrant well-being

Harsha Walia, in **Resisting Borders and Technologies of Violence**, critiques the state-imposed labels of "migrants" and "refugees" as tools for control that enforce power dynamics, determining inclusion and exclusion at borders [25]. These classifications, when paired with technology, create "selective inclusions" and "mass expulsions," impacting migrants' border experiences. For instance, in 2020, the UK Home Office's visa streaming algorithm was challenged by Foxglove and JCWI for disproportionately targeting specific nationalities based on flawed historical data. Introduced in 2015, the system perpetuated systemic discrimination, leading to its eventual removal under legal pressure, though concerns about the fairness of newer systems remain¹. Similarly, the UK's post-Brexit digital-only ID system, which disproportionately affects vulnerable groups like the elderly and digitally illiterate, raises significant concerns about accessibility and exclusion. These systems exacerbate the struggles of immigrants, who often face social, emotional, and psychological challenges during resettlement [7]. Institutional frameworks and societal attitudes play a crucial role in shaping their integration, yet insufficient legal protections and stigmatization amplify their vulnerability, leading to long-term mental health issues [14]. Border technologies, while aimed at enhancing security, frequently contribute to stress and anxiety among immigrants, forcing them to continuously prove their legitimacy [15]. This exacerbates inequalities and highlights the need for more humane, equitable immigration policies that prioritize well-being, dignity, and autonomy [8]. A shift toward such policies would enhance immigrants' quality of life and promote social cohesion [7].

¹<https://www.bbc.co.uk/news/technology-53650758>

3 RESEARCH GOALS & SUMMARY

This research seeks to examine the intersection of privacy, security, and well-being for immigrants subjected to AI-driven border technologies. Specifically, it aims to uncover how these technologies influence immigrants' privacy rights, mental health, and overall well-being. A key aspect of the research is its focus on Human-Computer Interaction (HCI) principles, which are applied to critically assess the design and functionality of these technologies. The study utilizes a decolonial theoretical framework to challenge dominant, state-driven narratives that often marginalize immigrant populations and reinforce systemic inequalities. This approach highlights the ethical and social dimensions of technology in border management, questioning how these systems perpetuate exclusion and control. So far, the research has employed semi-structured interviews with immigrants at UK borders to capture their lived experiences and perspectives on the technological complexities of border security systems. These interviews provide crucial insights into the human impact of AI technologies, offering a deeper understanding of how privacy, security, and well-being are entangled in the immigrant experience.

4 CURRENT WORK

Below I highlight my current ongoing work related to my research

4.1 Understanding immigrants' perceptions of border technologies

To answer RQ1, as part of my summer project, I conducted a study that revealed significant privacy and security concerns among immigrants regarding the use of AI-driven border technologies. I conducted my study in the UK, in association with a noted NGO (non government organization) in the country that provided me with access to immigrant participants and case workers who were willing to contribute to my research. Having conducted a thorough literature review, and understanding the gaps in current research, my investigation was primarily driven by two research questions: a) *How do immigrant populations perceive border control and security technologies and* b) *What impact do these technologies have on the emotional well-being of immigrant populations?* Many participants (both immigrants and case workers) voiced fears about how their data had been collected, stored, and potentially misused, particularly in cases like the discriminatory visa streaming algorithm, which reinforced biases against specific nationalities (Patel and Foxglove, 2020). A recurrent theme was the lack of transparency and consent surrounding how personal information was handled, with many feeling that their privacy was routinely violated by the constant surveillance at borders. Instead of providing a sense of security, these technologies often heightened feelings of vulnerability and state control, revealing a need for more transparent and privacy-respecting systems. Additionally, the constant surveillance and pressure to continually prove their legitimacy intensified feelings of stress, anxiety, and alienation. Borrowing from the statements of the case workers my study discovered that these systems, far from facilitating inclusion, deepened feelings of exclusion and insecurity, which directly undermined immigrant well-being. The essence of my findings narrated the importance of

developing more humane approaches to border management, prioritizing the dignity and autonomy of immigrants. This included the development of these technologies using a participatory approach wherein not only immigrant populations but also their care workers and protection personnel could also contribute their insights and opinions. Simultaneously, my research also advocated for the necessary training of security professionals at the borders when it comes to immigrants who come from extreme vulnerable and marginalized backgrounds, who may be in need of an increasing amount of human assurance and assistance that is to be reflected in the behavioral approach of these professionals. Finally, the work also suggested the need to stress of decolonizing border security technologies from a responsible, community centric and trustworthy lens especially with the ubiquitous advancement and use of artificial intelligence within these and upcoming systems. I am in the process of refining this research further for submission at a human centred security focused conference.

4.2 Studying security practitioners' approach towards developing and implementing border control technologies

This research intends on answering RQ2 and focuses on security practitioners involved in the design and implementation of border security systems, particularly as it relates to their thought processes regarding responsible technology development. I aim to investigate how these practitioners conceptualize responsibility within border technologies. The core objectives of this research are guided by the following questions: a) *What level of awareness do security practitioners possess concerning responsible border technologies* b) *How do they perceive current responsibility frameworks in technology development and design?* and c) *In what ways can their insights contribute to the development of improved frameworks for responsible technology development?* This exploration is essential as it seeks to illuminate their perspectives on the efficacy of decolonial approaches. To support this investigation, I am actively networking with security practitioners who operate at the borders, and border technology development professionals. I will be pursuing this research using a mixed methods approach wherein I will be using a survey questionnaire to gauge the level of awareness about responsible border technologies to answer the first research question of this investigation. The data from the survey will be analyzed using quantitative processes and insights from the same would be graphically visualized. After that, I will proceed towards carrying on semi-structured interviews with these two categories of practitioners. The interview protocol for this research, although will be developed from scratch, will be inspired by the foundations laid by scholars such as Botta [5] and Haney [12]. The interview data will undergo inductive coding and be thematically analyzed. The findings from this study are anticipated to be organized for publication at a conference relevant to the domains of human-computer interaction and human-centered artificial intelligence.

5 RESEARCH CONTRIBUTIONS AND NEXT STEPS

My study aims to apply a human-centred and critical computing lens to unravel the complexities of technology systems in border security. Fundamentals, such as usability, accessibility, and transparency, provide a important basis for evaluating the opaque nature of technology enabled decision-making systems for border control and monitoring purposes. The empirical approach of my research I believe will help in identifying how the systems' design and implementation often exacerbates existing inequalities, particularly by reinforcing biases embedded at the algorithmic and design levels [19]. By incorporating both qualitative and quantitative mechanisms into the evaluation of these technologies, the research could uncover ways to design more equitable and user-centered systems that could better serve marginalized groups for immigration purposes. I consider the decolonial framework that this study seeks to establish to be instrumental in challenging rather than perpetuating systemic power imbalances [16] when it comes to border control technologies. For the initial phase, the focus of my research was restricted to physical border technologies and human checks. However, future research will expand to include E-visas and other biometric or digital documentation systems that play an increasingly important role in border management finally culminating in a responsible and decolonial model of creating border control technologies to answer RQ3

6 BIO AND DISSERTATION STATUS

I am a PhD student at Royal Holloway, University of London, affiliated with the UKRI Centre for Doctoral Training (CDT) in **Cybersecurity and the Everyday** within the Department of Electronic Physical and Mathematical Sciences. My advisors are Prof. Rikke Bjerg Jensen and Prof. Maryam Mehrnezhad. I have completed the mandatory training session provided by the CDT, where my summer project marked the culmination of this training and facilitated my transition into further empirical research, as previously discussed. My PhD is generously supported by the UKRI CDT Scholarship. Additionally, I am a visiting research intern at the Max Planck Institute for Security and Privacy in Bochum, contributing to the Technology Harm Prevention Group. I plan to submit my final thesis in 2027, after which I aspire to secure research positions within the academic or industrial spaces.

References

- [1] Nina Amelung and Vasilis Galis. 2023. Border control technologies: introduction. *Sci. Cult. (Lond.)* 32, 3 (July 2023), 323–343.
- [2] Claudia Aradau. 2023. Algorithmic security and conflict in a datafied world. In *Digital International Relations*. Routledge, London, 177–197.
- [3] Claudia Aradau and Lucrezia Canzutti. 2022. Asylum, Borders, and the Politics of Violence: From Suspicion to Cruelty. *Global Studies Quarterly* 2, 2 (Jan. 2022), 1–11. <https://doi.org/10.1093/isagsq/ksab041>
- [4] Christina Boswell and James Besse. 2023. The strange resilience of the UK e-Borders programme: Technology hype, failure and lock-in in border control. *Secur. Dialogue* 54, 4 (Aug. 2023), 395–413.
- [5] David Botta, Rodrigo Werlinger, André Gagné, Konstantin Beznosov, Lee Iverson, Sidney Fels, and Brian Fisher. 2007. Towards understanding IT security professionals and their tools. In *Proceedings of the 3rd Symposium on Usable Privacy and Security* (Pittsburgh, Pennsylvania, USA) (SOUPS '07). Association for Computing Machinery, New York, NY, USA, 100–111. <https://doi.org/10.1145/1280680.1280693>
- [6] Lucrezia Canzutti and Claudia Aradau. 2024. Collecting, assembling, ordering: Border politics and the invisible data work of asylum. *Environ. Plan. C Politics Space* 42, 7 (Nov. 2024), 1242–1259.
- [7] Jiaee Cheong, Micol Spitale, and Hatice Gunes. 2023. “it’s not fair!” – fairness for a small dataset of multi-modal dyadic mental well-being coaching. In *2023 11th International Conference on Affective Computing and Intelligent Interaction (ACII)*. IEEE, Cambridge, MA, USA, 1–8.
- [8] Erzsébet Csatlós. 2024. Prospective implementation of ai for enhancing European (in)security: Challenges in reasoning of automated travel authorization decisions. *Comput. Law Secur. Rep.* 54, 105995 (Sept. 2024), 105995.
- [9] Virginia Dignum. 2019. *Responsible artificial intelligence* (1 ed.). Springer Nature, Cham, Switzerland.
- [10] Niall Docherty and Asia J. Biega. 2022. (Re)Politicizing Digital Well-Being: Beyond User Engagements. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 573, 13 pages. <https://doi.org/10.1145/3491102.3501857>
- [11] Tamy Guberek, Allison McDonald, Sylvia Simioni, Abraham H. Mhaidli, Kentaro Toyama, and Florian Schaub. 2018. Keeping a Low Profile? Technology, Risk and Privacy among Undocumented Immigrants. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3173574.3173688>
- [12] Julie M. Haney, Clyburn Cunningham IV, and Susanne M. Furman. 2024. Towards Bridging the Research-Practice Gap: Understanding Researcher-Practitioner Interactions and Challenges in Human-Centered Cybersecurity. In *Twentieth Symposium on Usable Privacy and Security (SOUPS 2024)*. USENIX Association, Philadelphia, PA, 567–586. <https://www.usenix.org/conference/soups2024/presentation/haney>
- [13] Lise Endregard Hemat. 2022. Trends in the Digitalisation of EU Borders: How Experimentations with AI for Border Control Treat Migrants as Test Subjects. PRIO Policy Brief, 15.
- [14] Christian Kerr and Nick Watts. 2022. Against a bitter tide: How a small UK charity operationalises dissent to challenge the “hostile environment” for migrant children and families. *Aotearoa N. Z. Soc. Work* 34, 3 (Sept. 2022), 61–73.
- [15] Younus Khan, Elizabeth Caldwell, and Jamie P Halsall. 2017. The missing link in the chain. *Illn. Crises Loss* 25, 4 (Oct. 2017), 323–339.
- [16] Shakir Mohamed, Marie-Therese Png, and William Isaac. 2020. Decolonial AI: Decolonial theory as sociotechnical foresight in artificial intelligence. *Philos. Technol.* 33, 4 (Dec. 2020), 659–684.
- [17] Sarah Perret and Claudia Aradau. 2024. Drawing data together: Inscriptions, asylum, and scripts of security. *Sci. Technol. Human Values* 49, 4 (July 2024), 739–764.
- [18] Christopher W Reynolds, Brendon Tucker, Samuel Bishop, Sarah Draugelis, Michele Heisler, and Amir M Mohareb. 2024. Health effects and user perceptions of the US Customs and Border Patrol One™ mobile application: A qualitative analysis among asylum seekers at the Mexico-US border. *J. Migr. Health* 10, 100265 (Aug. 2024), 100265.
- [19] Javier Sánchez-Monedero and Lina Dencik. 2022. The politics of deceptive borders: ‘biomarkers of deceit’ and the case of iBorderCtrl. *Inf. Commun. Soc.* 25, 3 (Feb. 2022), 413–430.
- [20] Samuel Singler. 2018. The Role of Technology in the Criminalization of Migration. <https://blogs.law.ox.ac.uk/research-subject-groups/centre-criminology/centreborder-criminologies/blog/2018/11/role-technology>. Accessed: 2024-10-22.
- [21] Wally Smith, Greg Wadley, Sarah Webber, Bernd Ploderer, and Reeva Lederman. 2014. Unbounding the interaction design problem: the contribution of HCI in three interventions for well-being. In *Proceedings of the 26th Australian Computer-Human Interaction Conference on Designing Futures: The Future of Design* (Sydney, New South Wales, Australia) (OzCHI '14). Association for Computing Machinery, New York, NY, USA, 392–395. <https://doi.org/10.1145/2686612.2686672>
- [22] Özgün E Topak. 2023. Drones: Robot eyes on racialized migrant bodies. *Int. Migr.* 61, 5 (Oct. 2023), 313–317.
- [23] Ana Valdivia, Claudia Aradau, Tobias Blanke, and Sarah Perret. 2022. Neither opaque nor transparent: A transdisciplinary methodology to investigate datafication at the EU borders. *Big Data Soc.* 9, 2 (July 2022), 1–17.
- [24] Bastian A Vollmer. 2019. The paradox of border security – an example from the UK. *Polit. Geogr.* 71 (May 2019), 1–9.
- [25] Harsha Walia. 2021. *Border and rule*. Haymarket Books, Chicago, IL.